



ME 342A: SYSTEM DESIGN AND ANALYSIS

UNIT 2 PROCEDURE FOR ENGINEERING A SYSTEM

Learning outcomes

- **System Objective**
- **Formulation of Objective Criteria**
- **Development of System Alternatives**



1 System Design Phases

A system engineer must design a system in the following phases in sequence of order:

1. Understanding the Problem
2. Considering the Alternative Solution
3. Choosing the most Optimum System
4. Synthesis of the System
5. Updating Equipment Characteristics and Data
6. Testing the System and Refinement
7. Application and Operation.

Phase 1 Understanding the Problem

In first phase a system engineer must understand the problem and requires the determination of the overall system requirements and objectives. While determining the system requirements, system engineer must consider the different factors like performance, reliability schedule, cost, maintainability power consumption, life expectancy and operating requirements and then give due weightage to each of these functions.

Phase 2 Considering the Alternative Solution

The next step is to consider alternative solutions. System engineer must be able to find out as many solutions as possible, as the more solutions are considered at initial level; greater will be the probability of getting a good system at final level. Then most promising and potentially useful solutions are given further consideration. Technical and financial feasibility study is conducted on these selected solutions. Generally, the promising solutions are reduced to three for conducting the feasible study.

Phase 3 Choosing the most Optimum System

In the next step, one solution is finally selected which is best among all the solutions. This selection be made by using the correct and reliable data and information so that correct decision be made. Accuracy in characteristics and



data at this level is important.

Phase 4 Synthesis of the System

This consists of the complete theoretical and physical design of the system. In this, effects of environment or internal forces on the stability of the system are determined. Socioeconomic conditions at the time when the product is expected to be put are also considered.

Phase 5 Updating Equipment Characteristics and Data

For further evolution of the overall system more emphasis is paid on the actual conditions and the system is further improved accordingly. For this purpose, feedback method is extremely useful. In feedback method, the information from each step is feedback to the previous steps for modification. This is repetitive in nature.

Phase 6 Testing the System and Refinement

Experimental models are then tested to check the entire ideas. The component and prototypes are also tested as need arises. These tests data are the basis for redesign and refinement and the process is continued till a process design is accomplished as per requirements.

Phase 7 Application and Operation

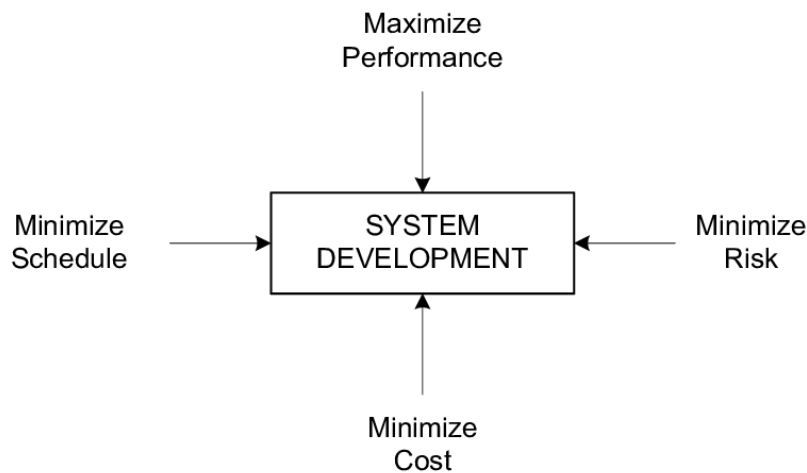
In the end the system design is applied for planning the production, distribution of goods etc. After getting the green signal from all corners the operation from business point of view is started.

2 Defining System Objective

The objectives are the specific results that a system aims to achieve within a time frame and with available resources. In general, objectives are more specific and easier to measure than goals. Objectives are basic tools that underlie all planning and strategic activities of a system. They serve as the basis for creating policy and evaluating performance. The objectives can be used in project planning for business, government, non-profit organizations, and even for personal use (for



example, in resumes to describe the exact position a jobseeker wants). A system may have one objective, many parallel objectives, or several objectives that must be achieved sequentially. To produce the most benefit, objectives must be defined early in the System Development Life Cycle (SDLC).



A system objective describes the desired results of a project, which often includes a tangible item. An objective is specific and measurable, and must meet time, budget, and quality constraints. System objectives are neutral (bias free), relating to, or based on verifiable evidence or facts instead of on attitude, belief, or opinion. The characteristics of system objectives are: Specific, Important, Measurable, Practical, Realistic, Feasible, Evaluable. The objectives should be **SMART**

S	Specific Make targets specific and precise
M	Measurable Make goals measurable by using benchmarks
A	Acceptable and Ambitious Goals must be acceptable to you and the stakeholders
R	Realistic and Relevant Goals must be achievable with available resources
T	Time-bound Goals have a clear start and end date

***Example: Financial planning objectives***

An important objective of personal financial planning is to ensure that unnecessary expenses are eliminated and that necessary expenditures are dealt with in a timely manner, so as not to generate late fees or other penalties.

Example: Manufacturing industry objectives**1. Reduce Cost**

Reducing manufacturing cost by a certain percentage over a given period acts as a good quality system objective for most manufacturing companies. Reducing manufacturing cost meets all three requirements of a solid quality system objective. First, the objective focuses on the customer. Cost reductions lead to increased profitability for the company. If a company chooses it can return some of this profitability back to the customer in the form of lower prices. This type of cost reduction also meets the second requirement for a good quality system objective - a company must measure the objective. In other words, the company must know its current performance and can track and monitor its future performance. This goal also meets the third requirement that states the quality objective must match the company's vision. Many manufacturing companies seek to reduce their manufacturing costs as part of their strategic vision.

2. Increase Service Levels

An increase in service levels generally leads to an increase in revenue. Service level defines how well a company meets the demands of the consumer. A company that has an 80 percent service level meets the demands of its customers 80 percent of the time. Service level equates to competitive edge. Typically, consumers give their business to the company that provides a better level of service. As a quality objective an increase in service level meets all three requirements. First, the core focus lies on the customer. Second, a company can measure the results of this objective. For example, a shipping company can document how many correct packages get delivered to the right customer at the right time. After implementation of the quality objective the company can



compare previous results with future results. It also meets the final requirement. Not too many companies would say increasing their level of service is not a part of their corporate vision.

3. Continual Product Improvement

Many companies, large and small, that produce a product desire to continually improve that product. This appears evident with the many “new and improved” versions of products that hit the shelves every year. Even the largest soft drink manufacturer in the world, Coca-Cola, tried to improve on a classic formula. As a quality system, objective continual product improvement does not always mean a new and improved product. Most often it means continually refining product quality and process efficiencies. This type of quality objective meets all three quality objective requirements. It is a customer-focused objective. Improved product quality benefits the consumer. Feedback from consumers and product testing panels provides measurement of the objective. Reduced complaints and product returns can be used as the measurement indicator. Most manufacturing companies make continual product improvement part of their strategic vision.

3 Formulation of Objective Criteria

System Analyst assesses what the future needs of the system will be and what changes should be considered to meet these needs. The objective criteria are formulated to select the best system that meets performance requirements. It comprises of identification, description, and evaluation of alternative systems and finally selection of the best system. This involves three steps:

1. Statement of constraints.
2. Identification of specific system objectives.
3. Description of outputs.

3.1 Statement of constraints

Constraints are factors that limit the solution of the problem. Some constraints are identified during the initial investigation and are discussed with the user.



There are general constraints that might have a bearing on the required performance of a candidate system.

Let us consider safe deposit billing system to illustrate these points. The current billing system and the department handling billing and customer payments face problems. The result of the fact-finding phase of the initial investigation revealed the following general constraints:

1. The president views safe deposit billing as a low priority. He has a false impression that computers are not needed if customers can have access to their boxes.
2. The senior vice president is worried that a billing system might require the transfer of safe deposit staff to other departments. Considering level of unemployment and the cost of retraining, a candidate system must do more than produce reports.
3. The accounting department has been pushing for installing a computer-based general ledger application for months. The vice president of operations, bogged down with auditing and operations problems, continued to shelve the request.
4. Management has a limited knowledge of computers, although it has several applications on the computer: checking and savings, instalment loans, commercial loans, and trusts. The president is interested in “the bottom line” of the financial statement, is traditionally reluctant to spend money on computers.
5. Safe deposit, while doing better than breaking even, is not projected to grow at fast rate.
6. If an online system is to be installed, it must interface with the existing checking/savings application to allow for the automatic payment of box rentals.
7. A proposed design must be compatible with the bank’s computer system.

3.2 Identification of specific system objectives

Once the constraints are spelled out, the analyst proceeds to identify the system’s specific performance objectives. They are derived from the general objectives specified in the project directive at the end of the initial investigation. The steps



are to state the system's benefits and then translate them into measurable objectives. For the case discussed above, the candidate system's anticipated benefits are as follows:

1. Improved collection schedule.
2. Cost reduction.
3. Physical space reduction.
4. Improved customer service.

Each benefit is analysed and translated into measurable objectives.

1. Collection is improved by billing 30 days in advance of the box renewal data, and one more notice is sent within two weeks. It also improves the account receivables payment "float."
2. Cost reduction is realized by reducing the payroll by two employees. The new online billing system requires less than two hours of labour per day, compared with six hours under the current system.
3. Placing the microcomputer in the place of one of the four existing desks reduces physical space requirements. The remaining desks are removed, allowing an extra cubicle for customer use.
4. Placing master cards and rental information online improve customer service, thus reducing the waiting time of entry from 3 minutes to 30 seconds.

These objectives are effective in comparing the performance of the candidate system with that of the current system. The information-oriented flowchart, input/output analysis sheet and data flow diagram produced in the initial investigation lead to the conclusions that (1) the current system is inefficient and (2) a new online, terminal-oriented system would be the solution. This conclusion was reflected in the general project directive submitted to the user for approval. This information is used as a basis for preparing specific objectives for the candidate system:

1. To establish a billing system with six five-day cycles per month.



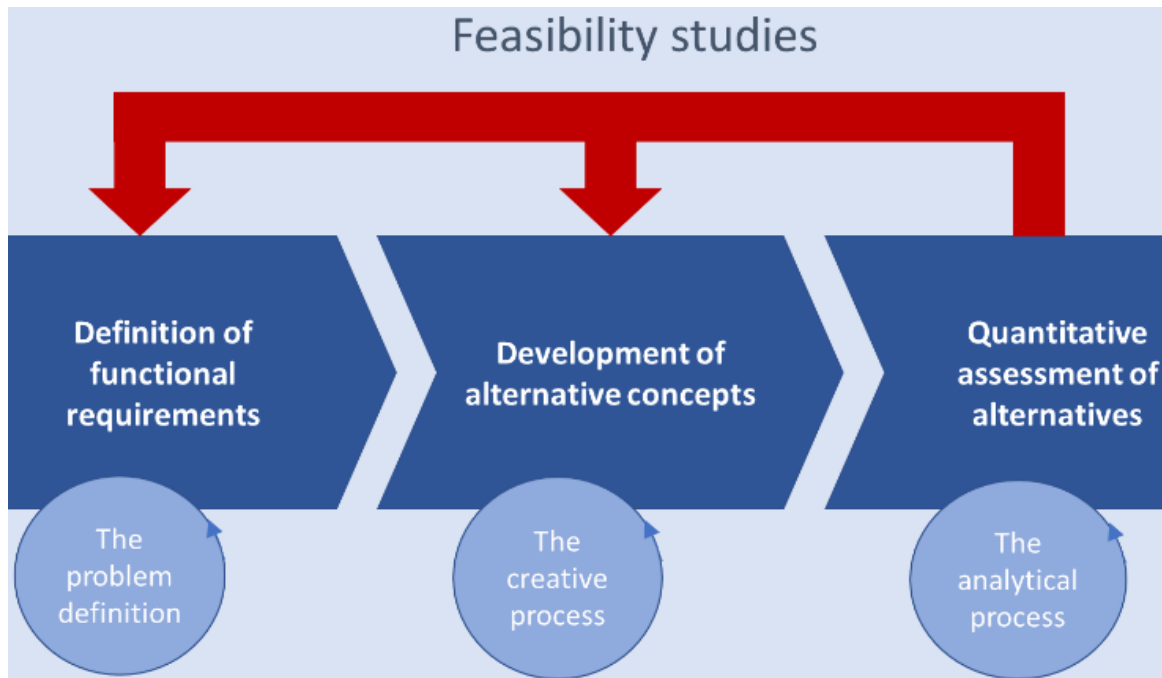
2. To mail customers no later than the close of the billing cycle and no later than 25 days prior to the box renewal date.
3. To mail customers a reminder two weeks after the initial statement for box renewal.
4. To speed collections and reduce the “float”.
5. To examine the availability of boxes by size, rental fees, and location.
6. To evaluate the ratio of always rented to available boxes.
7. To produce periodic reports to management on the performance of the safe deposit department.

3.3 Description of output

A final step in system performance definition is describing the output required by the user. An actual sketch of the format and contents of the reports (layout) as well as a specification of the media used, their frequency and the size and number of copies required are prepared at this point. Specifying exactly what the output will look like leads to an estimate of the computer storage requirements that form the basis for the file design to be undertaken in the design phase of the life cycle. The analyst is now ready to evaluate the feasibility of candidate systems to produce these outputs.

3.4 Feasibility study

Many feasibility studies are disillusioning for both users and analysts. First, the study often presupposes that when the feasibility document is being prepared, the analyst can evaluate solutions. Second most studies tend to overlook the confusion inherent in system development-the constraints and the assumed attitudes.



If the feasibility study is to serve as decision document, it must answer three key questions:

1. Is there a new and better way to do the job that will benefit the user?
2. What are the costs and savings of the alternative (s)?
3. What is recommended?

The most successful system projects are not necessarily the biggest or most visible in a business but rather those that truly meets user expectations. More projects fail because of inflated expectations than for any other reason.

Feasibility Considerations

Three key considerations are involved in the feasibility analysis: economic, technical, and behavioural. Let us briefly review each consideration and how it relates to the systems effort.

Economic Feasibility

Economic analysis is the most frequently used method for evaluating the effectiveness of a candidate system. More commonly known as cost/benefit analysis, the procedure is to determine the benefits and savings that are expected

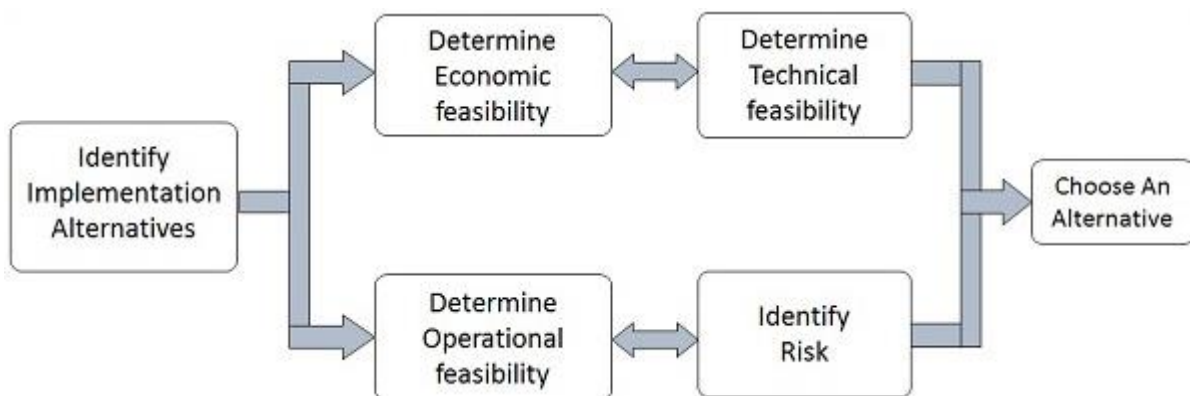
from a candidate system and compare them with costs. If benefits outweigh costs, then the decision is made to design and implement the system. Otherwise, further justification or alterations in the proposed system will have to be made if it is to have a chance of being approved. This is an ongoing effort that improves in accuracy at each phase of the system life cycle.

Technical Feasibility

Technical feasibility centers around the existing system (hardware, software, etc.) and to what extent it can support the proposed addition. For example, if the current computer is operating at 80 percent capacity - an arbitrary ceiling - then running another application could overload the system or require additional hardware. This involves financial considerations to accommodate technical enhancements. If the budget is a serious constraint, then the project is judged not feasible.

Behavioural Feasibility

People are inherently resistant to change, and computers have been known to facilitate change. An estimate should be made of how strong a reaction the user staff is likely to have toward the development of a computerized system. It is common knowledge that computer installations have something to do with turnover, transfers, retraining and changes in employee job status. Therefore, it is understandable that the introduction of a candidate system requires special effort to educate, sell and train the staff on new ways of conducting business.





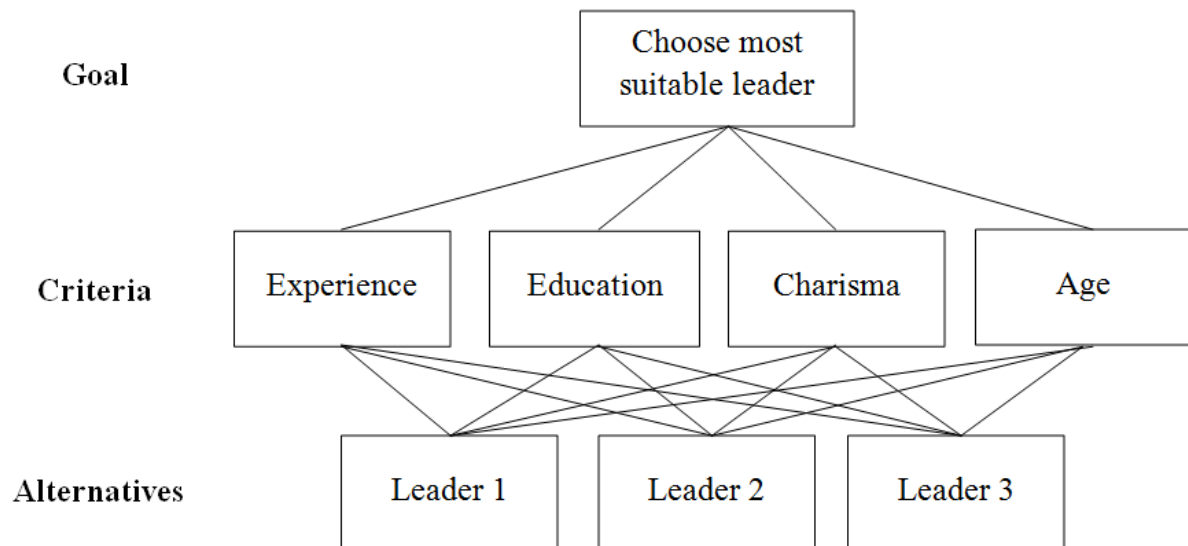
In safe deposit example, three employees are more than 50 years old and have been with the bank over 14 years, four of which have been in safe deposit. The remaining two employees are in their early thirties. They joined safe deposit about two years before the study. Based on data gathered from extensive interviews, the younger employees want the programmable aspects of safe deposit (essentially billing) put on a computer. Two of the three older employees have voiced resistance to the idea. Their view is that billing is no problem. The main emphasis is customer service – personal contact with customers. The decision in this case was to go ahead and pursue the project.

Steps in Feasibility Analysis

Feasibility analysis involves eight steps:

1. Form a project team and appoint a project leader.
2. Prepare system flowcharts.
3. Enumerate potential candidate systems.
4. Describe and identify characteristics of candidate systems.
5. Determine and evaluate performance and cost effectiveness of each candidate system.
6. Weight system performance and cost data.
7. Select the best candidate system.
8. Prepare and report final project directive to management.

3.5 Criteria for Selection of System Alternatives



Prior to selecting the system alternative, the management must set up criteria for selection. Selection criteria fall into the categories described below:

Reliability

It is the probability that the system will work for a specified period without a failure, weighted by the cost to the user of each failure encountered. It relates to the ease of recovery and ability to give consistent results. Reliability is particularly important to the professional user. For example, a pharmacist relies on past files on patients when filling prescriptions. Information accuracy is crucial. Hardware may become inoperative because of design errors, manufacturing errors, or deterioration caused by heat, humidity, friction, and the like. In contrast, software does not fail or wear out. Any reliability problems are attributable to errors introduced during the production process. Furthermore, whereas hardware failure is based largely on random failures, software reliability is based on predestined errors.

Functionality

It is a definition of the facilities, performance, and other factors that the user requires in the finished product. All such information comes from the user.

Capacity



Capacity refers to the capability of the system to handle the performance requirements in every working condition. All limitations should be checked.

Flexibility

It is a measure of the effort required to modify an existing system. The feature of flexibility is adaptability, which is a measure of the ease of extending the product.

Usability

This criterion refers to the effort required to operate a system or use a product/service. The additional points to be considered are portability and understandability. Portability refers to the ability of the system/device to be used in different working conditions. Understandability means that the purpose of the product is clear to the evaluator and that the package is clearly and simply written, is free of jargon, and contains sufficient references to readily available documents so that the user can comprehend the updates.

Security

It is a measure of the likelihood that a system's user can accidentally or intentionally access or destroy features of a system. A key question is how well can one control access of a system? Control provides system integrity.

Performance

It is a measure of the capacity of the system to do what it is expected to do. This criterion focuses on throughput, or how effectively it performs under the adverse conditions. Each system should be evaluated for acceptance based on various performance criteria. In many cases, the change could be costly and counterproductive.

Serviceability

This criterion focuses on documentation and system support. Complete documentation is critical for system performance. It includes a narrative description of the system, system logic and logic instructions. System support assures the user adequate technical support for software installation,



enhancements, and maintenance, the user should determine how much on – site technical assistance is provided by the vendor, especially during the first few weeks after the installation. The user expects on – site training and support as part of most commercial packages of systems. It is vital to inquire about the amount of training provided. The user may require training at several levels--clerical, operations, programming, and management.

Minimal costs

Cost is a major consideration in deciding between various system alternatives.

Cost – conscious users consider the following points:

1. Development and conversion costs.
2. Delivery schedule.
3. Cost and frequency of modifications.
4. Usable life span of the system.

4 Development of System Alternatives

System Analysts may recommend alternatives for improving the situation. Usually more than one strategy is possible. Time required to develop one alternative, compared with others, is the most critical issue. Costs and benefits are also important determinants. In the end, management, which will pay for and use the result, decides which alternative to accept.

System alternatives May include:

1. Upgrading or enhancing the existing system
2. Transferring a system or components from another State
3. Integrating with other systems
4. Developing a new system from the ground up

Alternative analysis predicts the changes to existing systems and processes along with organizational effect upon personnel and skill requirements. It requires the data conversion activities that analyze the operational impacts on existing procedures, user relationships, system support plans, resource effects during



development and operations. One can estimate the cost to operate the new system once implemented.

Alternatives analysis procedure

- a. Perform a gap analysis of program functional and technical requirements
- b. Analyze how many changes would be needed for each alternative to meet your needs
- c. Define your goal to meet your needs for a more efficient process

4.1 Need of system alternatives

A supervisor may want to investigate the system flow in purchasing, or a bank president has been getting complaints about the long lines. This need leads to a preliminary survey or an initial investigation to determine whether an alternative system can solve the problem. It entails looking into the duplication of effort, bottlenecks, inefficient existing procedures, or whether parts of the existing system would be candidates for computerization.

Example 1: Advertising

Objective is to develop a system to know: “How does the running of a TV advertising campaign designed to boost the image of a company affects its share price?”

Example 2: Job Recruitment via Internet

Objective is to develop a system to know: “How effective is recruiting new staff via the internet in comparison with traditional method?”

Entire group of alternatives is evaluated together to provide “big picture” view of proposed system. The key criteria used are:

- a) Strategic plan
- b) Economic feasibility
- c) Schedule and resource feasibility
- d) Technological feasibility
- e) Operational, organizational, and cultural feasibility



Generate a comprehensive set of alternative design strategies and select the one design strategy that is most likely to result in the desired information system.

4.2 Criteria for choosing best alternative

Cost - In-house versus purchased

Functionality - Mandatory, essential, and desired features

Vendor Support – Installation, Training, Technical Support

Viability of Vendor

Flexibility - Ease of customization

Documentation - User documentation, Technical documentation

Response Time

Ease of Installation

4.3 Alternatives types

a. Low-end

Provides all required functionality users demand with a system that is minimally different from the current system

b. High-end

Solves problem in question and provides many extra features users desire

c. Midrange

Compromise of features of high-end alternative with economical low-end alternative

4.4 Evaluation of alternatives

Once the alternative solutions have been developed, they must be evaluated so that the best solution can be identified. The goal of evaluation is to determine how well each alternative solution meets your business and personal requirements. These requirements are key characteristics and capabilities that you feel are necessary for your personal or business success.

Evaluation of alternatives – Example 1: Advertising



Objective is to develop a system to know: “How does the running of a TV advertising campaign designed to boost the image of a company affects its share price?”

The sales manager of a company might develop very specific requirements for solving the sales-related information problems of sales people. One can insist that any computer-based solution for sales force be reliable and easy to use. It is required that any proposed solution have low start-up costs, have minimal operating costs compared to present sales processing methods. Then develop evaluation criteria and determine how well each alternative solution meets these criteria. Develop criteria for such factors as start-up costs, operating costs, ease of use etc. Criteria may be ranked or weighted, based on their importance in meeting the requirements.

4.5 Selection criteria for alternatives

- a) Cost (development, procurement, support, product life cycle)
- b) Technical performance
- c) Complexity of the product component and product-related life-cycle processes
- d) Robustness to product operating and use conditions, operating modes, environments, and variations in product-related life-cycle processes
- e) Product expansion and growth
- f) Technology limitations
- g) Sensitivity to construction methods and materials
- h) Risk
- i) Evolution of requirements and technology
- j) Disposal
- k) Capabilities and limitations of end users and operators

4.6 Sources of alternatives

1. Re-define problem from others' points of view, including opponents of any



change.

2. Consider the ideal, then apply political, economic, and other constraints.
3. Start from generic, to modified, to custom-made alternatives.
4. Quick Surveys by telephone, fax, or e-mail, of people in policy issue network, public meetings, or hearings.
5. Literature review of professional and academic journals, government reports, collected proceedings from conferences.
6. Case studies of real-world experiences: why was the alternative adopted, what were the circumstances, what other alternatives were considered and discarded, how did it eventually work out, what modifications were made after implementation.
7. Passive collection and classification: keep a folder for collecting interesting solutions on a regular basis, even if no problem exists now, from clients, superiors, advocates, media, interest groups, etc. Then refer to the folder in emergencies.
8. Develop Typologies: identify all the types of persons likely to be affected by any alternative.
9. Use analogies to relate new problems with old problems
10. Brainstorming - can be oral, written, or electronic. Brainstorming has two phases, first a pure idea-generation phase, where no judgements are made about any ideas; and second, an evaluation and ranking phase, to help arrive at concrete solutions.
11. Feasible Manipulation - develops alternatives based on limited, moderate, or wide manipulation of possible activities.

Review Questions



1. Explain in details the various steps involved in system design.
2. Define system objective and discuss in details the SMART objectives.
3. Describe in details the formulation of system objectives criteria.
4. Explain the formulation of objectives criteria for a manufacturing industry.
5. Explain the importance of system alternatives and criteria for choosing the best alternative.
6. Elaborate System performance.
7. What is the importance of feasibility study? Explain economic, technical, and behavioral feasibility.
8. Write about the various criteria used to select the best solution among the alternate solutions for a system.
9. Describe in details the various sources of the alternative solutions.